

From the nominal cosh to the fitted cosh: what the minimiser absorbs, step by step

every intermediate carried; nothing skipped

The question is: if the true signal is the nominal response plus a deviation e_ω , and the minimiser fits a cosh to the sum, what is left over? The answer decides what a fit residual can and cannot tell us, so it is worth doing slowly. Sections 1–3 set up e_ω , Sec. 4 is the one piece of algebra everything turns on, and Secs. 5–8 apply it.

1 The nominal response, from the dynamics

Over the contact phase the linearised tip-over dynamics are

$$J_P \ddot{\varphi} = Wz_{CoM} \varphi + (M(t) - M_{\text{crit}}), \quad (1)$$

the sign on Wz_{CoM} being positive because past the balance point gravity *drives* the tip rather than restoring it. Measure time from the onset, $\tau = t - t_c$, so that $M(t) - M_{\text{crit}} = \dot{M}\tau$, and divide by J_P :

$$\ddot{\varphi} - C_2^2 \varphi = \frac{\dot{M}\tau}{J_P}, \quad C_2^2 \triangleq \frac{Wz_{CoM}}{J_P}. \quad (2)$$

Particular solution. Try $\varphi_p = \alpha\tau$. Then $\ddot{\varphi}_p = 0$ and (2) gives $-C_2^2 \alpha\tau = \dot{M}\tau/J_P$, so

$$\alpha = -\frac{\dot{M}}{J_P C_2^2} = -\frac{\dot{M}}{Wz_{CoM}}, \quad \text{i.e.} \quad \varphi_p = -\frac{\dot{M}}{Wz_{CoM}} \tau.$$

Homogeneous solution. $\ddot{\varphi}_h = C_2^2 \varphi_h$ gives $\varphi_h = A \cosh C_2 \tau + B \sinh C_2 \tau$. So

$$\varphi(\tau) = A \cosh C_2 \tau + B \sinh C_2 \tau - \frac{\dot{M}}{Wz_{CoM}} \tau. \quad (3)$$

Initial conditions. At the onset the vehicle is at rest at the balance point: $\varphi(0) = 0$ and $\dot{\varphi}(0) = 0$. The first gives $A = 0$. Differentiating, $\dot{\varphi} = BC_2 \cosh C_2 \tau - \dot{M}/Wz_{CoM}$, and at $\tau = 0$,

$$BC_2 - \frac{\dot{M}}{Wz_{CoM}} = 0 \quad \implies \quad B = \frac{\dot{M}}{Wz_{CoM} C_2}.$$

The rate. Substituting A and B and differentiating once,

$$\omega_{\text{nom}}(\tau) = \dot{\varphi} = \frac{\dot{M}}{Wz_{CoM}} (\cosh C_2 \tau - 1) = C_1 (\cosh C_2 \tau - 1), \quad C_1 \triangleq \frac{\dot{M}}{Wz_{CoM}} = K \dot{M}. \quad (4)$$

Two things to notice, because both matter later. The amplitude C_1 is *not free*: it is fixed by the ramp rate and the rig constant K . And $\omega_{\text{nom}}(0) = 0$, so the curve leaves the onset flat.

2 The deviation, when an unmodelled moment acts

Let $\rho(\tau)$ be a moment the model does not contain. The true tilt obeys (1) with ρ added:

$$J_P \ddot{\varphi} = W z_{CoM} \varphi + \dot{M} \tau + \rho(\tau).$$

Write $\varphi = \varphi_{\text{nom}} + e_\varphi$ and subtract (2). Everything the two have in common cancels, leaving

$$\ddot{e}_\varphi - C_2^2 e_\varphi = \frac{\rho(\tau)}{J_P}, \quad e_\varphi(0) = \dot{e}_\varphi(0) = 0, \quad (5)$$

with zero initial conditions because the nominal solution already satisfies them.

Impulse response. For $\ddot{y} - C_2^2 y = \delta(\tau)$ with $y(0) = \dot{y}(0) = 0$ the solution is $h(\tau) = \sinh(C_2 \tau)/C_2$: it vanishes at $\tau = 0$, its derivative $\cosh C_2 \tau$ equals 1 there, which is the unit jump the impulse delivers, and it satisfies the homogeneous equation away from the origin.

Duhamel. Hence

$$e_\varphi(\tau) = \frac{1}{J_P} \int_0^\tau \frac{\sinh C_2(\tau - s)}{C_2} \rho(s) ds. \quad (6)$$

Differentiate with respect to τ . The Leibniz rule gives a boundary term plus an integral,

$$\dot{e}_\varphi = \underbrace{\frac{\sinh C_2(\tau - \tau)}{C_2 J_P} \rho(\tau)}_{=0} + \frac{1}{J_P} \int_0^\tau \cosh C_2(\tau - s) \rho(s) ds,$$

and the boundary term vanishes because $\sinh 0 = 0$. So

$$e_\omega(\tau) = \frac{1}{J_P} \int_0^\tau \cosh(C_2(\tau - s)) \rho(s) ds. \quad (7)$$

3 Splitting the kernel

The addition formula $\cosh(u - v) = \cosh u \cosh v - \sinh u \sinh v$ applies to the kernel with $u = C_2 \tau$ and $v = C_2 s$:

$$\cosh C_2(\tau - s) = \cosh C_2 \tau \cosh C_2 s - \sinh C_2 \tau \sinh C_2 s.$$

The factors in τ do not depend on s , so they come out of the integral:

$$e_\omega(\tau) = \frac{1}{J_P} \left[\cosh C_2 \tau P(\tau) - \sinh C_2 \tau Q(\tau) \right], \quad (8)$$

$$P(\tau) \triangleq \int_0^\tau \cosh C_2 s \rho(s) ds, \quad Q(\tau) \triangleq \int_0^\tau \sinh C_2 s \rho(s) ds. \quad (9)$$

This is exact — no approximation yet. P and Q are the running moments of ρ against the two kernels; if ρ has died away by some time, they stop growing and become constants from there on.

4 Two hyperbolics make one

Claim. $a \cosh u + b \sinh u = R \cosh(u + \psi)$ with $R = \sqrt{a^2 - b^2}$ and $\tanh \psi = b/a$, valid when $|a| > |b|$.

Proof. Look for R, ψ with $a = R \cosh \psi$ and $b = R \sinh \psi$. Then

$$R \cosh(u + \psi) = R [\cosh u \cosh \psi + \sinh u \sinh \psi] = (R \cosh \psi) \cosh u + (R \sinh \psi) \sinh u = a \cosh u + b \sinh u,$$

which is the claim. It remains to check that such R, ψ exist. Squaring and subtracting,

$$a^2 - b^2 = R^2(\cosh^2 \psi - \sinh^2 \psi) = R^2,$$

so $R = \sqrt{a^2 - b^2}$, real exactly when $|a| > |b|$; and dividing, $b/a = \tanh \psi$, which has a unique solution ψ because $\tanh$ is a bijection from $\mathbb{R}$ to $(-1, 1)$ and $|b/a| < 1$. $\square$

Read it as a time shift. Since $u = C_2\tau$,

$$R \cosh(C_2\tau + \psi) = R \cosh\left(C_2(\tau + \psi/C_2)\right),$$

so adding a $\sinh$ to a $\cosh$ does not produce a new shape at all: it produces the *same* cosh, with the amplitude changed from a to R and the time origin moved by $-\psi/C_2$.

5 The measured signal is again a cosh

Add the nominal (4) and the deviation (8), and write b_0 for the true pre-onset level:

$$\begin{aligned} y(\tau) - b_0 &= \underbrace{C_1 \cosh C_2\tau - C_1}_{\text{nominal}} + \underbrace{\frac{P}{J_P} \cosh C_2\tau - \frac{Q}{J_P} \sinh C_2\tau}_{e_\omega} \\ &= \underbrace{\left(C_1 + \frac{P}{J_P}\right)}_a \cosh C_2\tau + \underbrace{\left(-\frac{Q}{J_P}\right)}_b \sinh C_2\tau - C_1. \end{aligned} \quad (10)$$

Wherever P and Q vary slowly enough to be treated as constants, Sec. 4 applies verbatim:

$$\boxed{y(\tau) - b_0 = R \cosh\left(C_2(\tau - \delta)\right) - C_1, \quad R = \sqrt{a^2 - b^2}, \quad \delta = -\frac{\psi}{C_2}, \quad \tanh \psi = \frac{b}{a}.} \quad (11)$$

The perturbed signal is exactly a cosh. Not approximately, not to leading order — exactly, with a different amplitude and a different onset.

6 What the minimiser can absorb

The estimator fits

$$\hat{\omega}(\tau) = \hat{C}_1 \left(\cosh(C_2(\tau - \hat{\delta})) - 1 \right) + C, \quad (12)$$

with C the baseline. Expand the shifted cosh by the addition formula:

$$\hat{\omega} = \hat{C}_1 \cosh(C_2\hat{\delta}) \cosh C_2\tau - \hat{C}_1 \sinh(C_2\hat{\delta}) \sinh C_2\tau - \hat{C}_1 + C. \quad (13)$$

Now compare (13) with the data (10), coefficient by coefficient. There are three independent shapes present — $\cosh C_2\tau$, $\sinh C_2\tau$ and the constant 1 — so three equations:

shape	data	fit
$\cosh C_2\tau$	$a = C_1 + P/J_P$	$\hat{C}_1 \cosh C_2\hat{\delta}$
$\sinh C_2\tau$	$b = -Q/J_P$	$-\hat{C}_1 \sinh C_2\hat{\delta}$
1	$-C_1 + b_0$	$-\hat{C}_1 + C$

Case A: amplitude, onset and baseline all free. Three unknowns ($\hat{C}_1, \hat{\delta}, C$), three equations, and they are solvable — take $\hat{C}_1 = R$, $\hat{\delta} = \delta$ from (11), and $C = b_0 - C_1 + R$. *The fit is exact and the residual is identically zero.* Not small: zero. A fit with both shape parameters free cannot see e_ω at all, because e_ω has moved the signal along the model family rather than off it.

Case B: the amplitude pinned, $\hat{C}_1 = C_1 = \dot{K}\dot{M}$. Now there are two unknowns for three equations. The sinh row fixes the onset,

$$\sinh C_2 \hat{\delta} = -\frac{b}{C_1} = \frac{Q}{J_P C_1}, \quad (14)$$

and the constant row fixes C . The cosh row is then left over, and what it cannot match is the residual:

$$\text{residual coefficient on } \cosh C_2 \tau = a - C_1 \cosh C_2 \hat{\delta} = C_1 + \frac{P}{J_P} - C_1 \sqrt{1 + \left(\frac{Q}{J_P C_1}\right)^2}, \quad (15)$$

using $\cosh = \sqrt{1 + \sinh^2}$ with (14). For a small deviation the square root is $1 + O(Q^2)$ and this collapses to

$$\boxed{\text{residual} \simeq \frac{P(\tau_{\text{end}})}{J_P} \cosh C_2 \tau + O(Q^2).} \quad (16)$$

So pinning the amplitude is what makes anything observable, and what survives is the cosh half of the deviation — the P half. The sinh half, the Q half, has gone entirely into the onset.

7 What went into the onset, and what it costs

Solve (14) for small $\hat{\delta}$, where $\sinh C_2 \hat{\delta} \simeq C_2 \hat{\delta}$:

$$\hat{\delta} \simeq \frac{Q(\tau_{\text{end}})}{J_P C_1 C_2}. \quad (17)$$

The identified threshold is the moment at the identified onset, and the moment ramps at $\dot{M}$, so a shift $\hat{\delta}$ costs $\Delta M_{\text{crit}} = \dot{M} \hat{\delta}$. Substituting $C_1 = \dot{M}/Wz_{CoM}$ and then $Wz_{CoM} = J_P C_2^2$,

$$\Delta M_{\text{crit}} = \dot{M} \cdot \frac{Q}{J_P C_2} \cdot \frac{Wz_{CoM}}{\dot{M}} = \frac{Q Wz_{CoM}}{J_P C_2} = \frac{Q J_P C_2^2}{J_P C_2} = C_2 Q(\tau_{\text{end}}) = C_2 \int_0^{\tau_{\text{end}}} \sinh(C_2 s) \rho(s) ds. \quad (18)$$

The ramp rate has cancelled twice over: once because $C_1 \propto \dot{M}$ and once because $\Delta M_{\text{crit}} = \dot{M} \hat{\delta}$. What reaches the threshold is a property of the rig and of ρ , not of how fast the run was driven.

The step from $\sinh C_2 \hat{\delta}$ to $C_2 \hat{\delta}$ is the only approximation in the section. Keeping the exact root $\hat{\delta} = \text{arcsinh}(z)/C_2$ with $z = Q/(J_P C_1)$ and expanding $\text{arcsinh } z = z - z^3/6 + \dots$, the term dropped is $\dot{M} z^3/(6C_2)$ — on a representative run 9.2×10^{-6} mNm against the 1.52 of the leading term, so it never matters here, but it is what a numerical check will find if the two forms are compared to machine precision (`analysis/absorption_check.py`).

A caveat on (18). It was obtained by matching coefficients, which is exact only when P and Q have stopped growing — that is, when ρ acts early and then dies. In general the estimator does least squares over the whole window rather than matching coefficients, and the onset it picks is

$$\hat{\delta} = \frac{\langle e_\omega, \chi \rangle}{\|\chi\|^2}, \quad \chi \triangleq \frac{\partial \hat{\omega}}{\partial t_c} = -C_1 C_2 \sinh C_2 \tau, \quad (19)$$

which is the projection of the deviation onto the one direction the onset can move it. The two agree in the early- ρ limit and differ otherwise; the manuscript uses the projection form throughout, and it is that form which produces the normalised weight w with $\int w = 1$.

8 Three consequences, and the numbers

(i) **The residual and the threshold error are complementary.** By construction the least-squares onset makes the residual orthogonal to χ . So the residual is everything *orthogonal* to the estimator’s span, and ΔM_{crit} is exactly the χ -projection the residual has thrown away. Neither carries any information about the other, and comparing a fit residual against a bound on ΔM_{crit} tests nothing.

(ii) **An impulse at the onset does not move the onset.** If ρ acts only at $\tau \approx 0$ then $Q = \int \sinh(C_2 s) \rho ds \approx 0$ because $\sinh 0 = 0$, while $P \approx \int \rho$. By (18) the threshold does not move at all; by (16) the whole thing shows up as an amplitude error. The later ρ acts, the more of it is sinh-weighted and the more it displaces the onset — which is why the manuscript’s Chebyshev step, which assumes ρ concentrated *late*, is the pessimistic one.

(iii) **Releasing C_1 would destroy the measurement.** Case A says the residual would be zero, so the fit would be free to wander along the family. Measured on these runs, releasing amplitude and onset together drives the onset to ± 150 ms — the limit imposed on the search — at every ramp rate, with the amplitude moving from -17 to $+44\%$. At $\dot{M} = 1.20$ N m/s, 150 ms of onset is 180 mN m of threshold, fifteen times the entire a priori budget.

quantity	what it is	measured
$P/(J_P C_1)$	the cosh half, (16)	1.7% median, 3.9% at p90
$C_2 Q$	the sinh half, (18)	0.049 mN m median, 0.93 worst
	against the a priori bound	12.04 mN m
onset if C_1 released	Case A	± 150 ms, i.e. 180 mN m
span projection, one run	what the estimator absorbs	$0.555 \rightarrow 0.010$ mN m

Reproduced by `analysis/amplitude_channel.py` for the first row, `analysis/error_budget.py` and `analysis/bound_validation_figure.py` for the rest.